Homework #7. Due on Thursday, Oct 22nd, 11:59pm on Canvas

Reading for this homework: Sections 3.2 and 4.2.

Plan for next week (Oct 19, 21): Determinants (continued), 4.3-4.5.

Problems:

Problem 1: Let $n \in \mathbb{N}$ and $A, B \in \mathrm{Mat}_n(F)$.

(a) Use what we proved about rank (but not determinants!) to show that if AB is invertible, then A and B are both invertible (as discussed in class, the converse is also true).

(b) Deduce from (a) that if $AB = I_n$, then A and B are invertible, $B = A^{-1}$ and $A = B^{-1}$. Thus, when we are proving that a matrix Y is the inverse of a matrix X using the definition of inverse, it suffices to check that either $XY = I_n$ or $YX = I_n$ (no need to check both).

Problem 2: Let V and W be finite-dimensional vector spaces over the same field F, let $n = \dim(V)$, $m = \dim(W)$, and choose ordered bases β of V and γ of W. Let $T : V \to W$ be a linear map. Prove that $\mathrm{rk}([T]_\beta^\gamma) = \mathrm{rk}(T)$. See a hint at the end of the assignment.

Problem 3: Read (and understand) Corollary 3 on page 158 which asserts that every invertible matrix can be written as a product of elementary matrices.

In Problems 4-6 you are asked to prove certain facts about determinants. In all these problems do not use any facts about determinants except the definition from class (the Leibniz formula).

Problem 4: Let $A = (a_{ij}) \in \mathrm{Mat}_n(F)$.

(a) Suppose that A is diagonal, that is, $a_{ij} = 0$ for all $i \neq j$. Prove that $\det(A) = a_{11}a_{22}\dots a_{nn}$.

(b) Now suppose that A is upper-triangular, that is, $a_{ij} = 0$ for all $i > j$. Again prove that $\det(A) = a_{11}a_{22}\dots a_{nn}$.

(c) Let A be upper-triangular, and suppose that $a_{kk} = 0$ for some k. Prove without using determinants that $\mathrm{rk}(A) < n$. **Hint:** Consider the first k columns of A.

Now recall some notations from Lecture 13. Given $n \in \mathbb{N}$, let $[n] = \{1, \dots, n\}$ and S_n the set of all permutations of $[n]$. Given $\sigma \in S_n$, let $\mathrm{Inv}(\sigma)$ be the set of inversions of σ and $\mathrm{inv}(\sigma) = |\mathrm{Inv}(\sigma)|$, the number of inversions. Recall that according to the definition from class, an inversion of σ is an unordered pair $\{i, j\}$ with $i, j \in [n]$ such that $i < j$ and $\sigma(i) > \sigma(j)$ or

$i > j$ and $\sigma(i) < \sigma(j)$ (there are different conventions regarding the precise definition of inversion, but they do not affect the number of inversions).

Problem 5: Let $n \in \mathbb{N}$.

(a) Prove that for any $\sigma \in S_n$ we have $\mathrm{inv}(\sigma^{-1}) = \mathrm{inv}(\sigma)$, that is, σ and σ^{-1} have the same number of inversions. Here σ^{-1} is the inverse permutation, that is, the inverse function of σ. **Hint:** Construct an explicit bijection between $\mathrm{Inv}(\sigma)$ and $\mathrm{Inv}(\sigma^{-1})$.

(b) Use (a) and the Leibniz formula to prove that $\det(A^t) = \det(A)$ for all $A \in \mathrm{Mat}_n(F)$. We will give a very different proof of this fact in class.

Problem 6: Suppose that $M = (m_{ij}) \in \mathrm{Mat}_n(F)$ has block-diagonal form:

$$M = \begin{pmatrix} A_{k\times k} & B_{k\times(n-k)} \\ 0_{(n-k)\times k} & C_{(n-k)\times(n-k)}. \end{pmatrix}$$

for some $1 \leq k \leq n-1$. (Equivalently, suppose that $m_{ij} = 0$ whenever $i > k$ and $j \leq k$). Prove that

$$\det(M) = \det(A)\det(C).$$

See a hint at the end of the assignment.

Problem 7: Let F be a field, $n \in \mathbb{N}$ and $A = (a_{ij}) \in \mathrm{Mat}_n(F)$.

(a) Prove that A is invertible if and only if its columns are linearly independent.

(b) Let V be a vector space over F with $\dim V = n < \infty$, and let $\beta = \{v_1, \ldots, v_n\}$ be a basis of V. Define the vectors $v_1', \ldots, v_n' \in V$ by $v_j' = \sum_{i=1}^n a_{ij} v_i$. Prove that $\{v_1', \ldots, v_n'\}$ are linearly dependent if and only if the columns of A are linearly dependent (this can be done by a direct calculation). Deduce from (a) that $\{v_1', \ldots, v_n'\}$ is a basis if and only if A is invertible.

(c) Deduce from (b) that for any invertible matrix $A \in \mathrm{Mat}_n(F)$ and any basis β of V there is an ordered basis β' of V such that $A = [I]_{\beta'}^{\beta}$ (where $I : V \to V$ is the identity map).

Hint for 2: Consider the linear maps $\Phi : V \to F^n$ and $\Psi : W \to F^m$ given by $\Phi(v) = [v]_\beta$ and $\Psi(w) = [w]_\gamma$. Let $A = [T]_\beta^\gamma$. Prove that $L_A\Phi = \Psi T$ as maps from V to F^m (where as usual L_A is the left multiplication by A) – this is actually a theorem from class, just stated using very different notations. Deduce that $\mathrm{rk}(L_A) = \mathrm{rk}(T)$ using what you know about Φ, Ψ and the general properties of rank.

Hint for 6: Let $S_{n,k}$ be the set of all $\sigma \in S_n$ such that $\sigma([k]) = [k]$ (this is equivalent to saying that the first k entries in the one line notation for σ are integers from 1 to k, in some order). First prove that if $\sigma \in S_n \setminus S_{n,k}$, then the product in the Leibniz formula for $\det(M)$ corresponding to σ is equal to 0.

Next given $\sigma \in S_{n,k}$, define $\sigma_1 \in S_k$ and $\sigma_2 \in S_{n-k}$ as follows: σ_1 is simply the restriction of σ to $[k]$, and σ_2 is defined by $\sigma_2(i) = \sigma(i+k) - k$ for all $i \in [n-k]$. The description in terms of the one line notation is even simpler – the one line notation for σ_1 is simply the first k entries in the one line notation for σ; the one line notation for σ_2 is obtained from the one line notation for σ by deleting the first k entries and subtracting k from the last $n-k$ entries.

Prove that the map $f : S_{n,k} \to S_k \times S_{n-k}$ given by $f(\sigma) = (\sigma_1, \sigma_2)$ is a bijection and $\mathrm{inv}(\sigma_1) + \mathrm{inv}(\sigma_2) = \mathrm{inv}(\sigma)$ (and hence $(-1)^\sigma = (-1)^{\sigma_1} \cdot (-1)^{\sigma_2}$).

Now combine all these facts to represent $\det(M)$ as a product of two factors and show that the factors are equal to $\det(B)$ and $\det(C)$, respectively.